THAT WHICH IS CLAIMED IS:

1. A method of detecting a target analyte and/or performing a spatial multiplexed assay, the method comprising:

providing a microplate comprising:

a plurality of wells arranged in a two-dimensional array, each of the wells comprising:

a bottom surface,

sidewalls extending from the bottom surface to form an open top;

at least two subwells in the bottom surface, the at least two subwells having sidewalls that extend below the bottom surface of the well, wherein each of the at least two subwells comprises a capture binding agent that is configured to bind to a target analyte, if present, in a sample;

adding a sample to the well such that the sample fluidically contacts each of the at least two subwells such that a target analyte, if present, binds to the capture binding agent in one or more of the at least two subwells, wherein an upconverting nanoparticle (UCNP) labeled conjugate in the sample binds with the target analyte, if present; and

detecting a label of the upconverting nanoparticle (UCNP) labeled conjugate at one or more of the at least two subwells to thereby determine whether the target analyte is present in the sample.

2. The method of Claim 1, wherein the capture binding agent comprises at least a first capture binding agent and a second capture binding agent in different ones of the at least two subwells.

3. The method of Claim 2, wherein the UCNP labeled conjugate comprises a first UCNP labeled conjugate and a second UCNP labeled conjugate that are configured to bind to corresponding target analytes bound to the first and second capture binding agents, respectively, in the at least two subwells.

4. The method of Claim 3, wherein a UCNP of the first and second UCNP labeled conjugates is the same UCNP, and detecting a label of the labeled conjugate at one or more of the at least two subwells to thereby determine whether the target analyte is present in the sample

comprises determining whether first and second target analytes are present based on which of the at least two subwells emit a signal from the UCNP labeled conjugate.

5. The method of Claim 4, wherein detecting a UNCP label of the UNCP labeled conjugate at one or more of the at least two subwells comprises impinging photons on the at least two subwells such that the UCNP labeled conjugate emits a signal.

6. The method of Claim 5, further comprising determining a concentration of the target analyte based on a signal strength of a signal from the UCNP labeled conjugate, wherein the signal has a signal strength that corresponds to a concentration of the target analyte, if present, in the sample.

7. The method of Claim 6, wherein at least one of the first and second capture binding agents comprises a control capture binding agent and other of the first and second capture binding agents comprises a test capture binding agent, wherein a signal strength of a UCNP labeled conjugate bound to the test capture binding agent relative to a signal strength of a UCNP labeled conjugate bound to the control capture binding agent corresponds to the concentration of the target analyte, if present, in the sample.

8. The method of Claim 6, wherein determining the concentration of the target analyte based on the signal strength of the signal from the UCNP labeled conjugate is based on an empirically-based model of actual experience.

9. The method of Claim 1, wherein the capture binding agent comprises a first half of a binding pair and the capture binding agent is bound to a surface of the at least two subwells via binding of the first half of the binding pair to a second half of the binding pair on the surface of the at least two subwells.

10. The method of Claim 1, wherein the upconverting nanoparticle (UCNP) labeled conjugate is present on a bottom and/or side of the well prior to adding the sample to the well.

11. The method of Claim 1, further comprising adding the upconverting nanoparticle (UCNP) labeled conjugate to the sample.

12. The method of claim 1, wherein the target analyte is human chorionic gonadotropin (hCG).

13. The method of claim 1, wherein the target analyte is placental growth factor (PlGF).

14. A microplate comprising:

a plurality of wells arranged in a two-dimensional array, each of the wells comprising:

a bottom surface,

sidewalls extending from the bottom surface to form an open top;

at least two subwells in the bottom surface, the at least two subwells having sidewalls that extend below the bottom surface of the well and being sized and configured such that a sample added into the well fluidically contacts each of the at least two subwells, wherein each of the at least two subwells comprises a capture binding agent that is configured to bind to a target analyte, if present, in the sample, which is further configured to bind with a upconverting nanoparticle (UCNP) labeled conjugate in the at least two subwells.

15. The microplate of Claim 14, wherein the capture binding agent comprises at least a first capture binding agent and a second capture binding agent in different ones of the at least two subwells.

16. The microplate of Claim 15, wherein the UCNP labeled conjugate comprises a first UCNP labeled conjugate and a second UCNP labeled conjugate that are configured to bind to corresponding target analytes bound to the first and second capture binding agents, respectively, in the at least two subwells.

17. The microplate of Claim 16, wherein a UCNP of the first and second UCNP labeled conjugates is the same, and the microplate is configured to be analyzed by a reader that determines whether the first and second target analytes are present based on which of the at least two subwells emit a signal from the UCNP.

18. The microplate of Claim 17, wherein the UNCP labeled conjugate at one or more of the at least two subwells is configured to be detected by impinging photons on the at least two subwells such that the UCNP labeled conjugate emits a signal.

19. The microplate of Claim 18, wherein the microplate is configured such that a concentration of the target analyte is determined based on a signal strength of a signal from the UCNP labeled conjugate, wherein the signal has a signal strength that corresponds to a concentration of the target analyte, if present, in the sample.

20. The microplate of Claim 19, wherein at least one of the first and second capture binding agents comprises a control capture binding agent and the other of the first and second capture binding agents comprises a test capture binding agent, wherein a signal strength of a UCNP labeled antibody bound to the test capture binding agent relative to a signal strength of a UCNP labeled antibody bound to the control capture binding agent corresponds to the concentration of the target analyte, if present, in the sample.

21. The microplate of Claim 20, wherein a concentration of the target analyte based on a signal strength of a signal from the UCNP labeled conjugate is based on an empirically-based model of actual experience.

22. The microplate of Claim 14, wherein the capture binding agent comprises a first half of a binding pair and the capture binding agent is bound to a surface of the at least two subwells via binding of the first half of the binding pair to a second half of the binding pair on the surface of the at least two subwells.

23. The microplate of Claim 16, wherein the upconverting nanoparticle (UCNP) labeled conjugate is present on a bottom and/or side of the well.